

Student Name:

Q1 : Choose the Correct answer

1. Which one of the following logic expression is incorrect?

A. $1 \oplus 1 \oplus 1 = 1$ ✓

B. $1 \oplus 1 \oplus 0 = 1$ ✗

C. $1 \oplus 0 \oplus 0 = 1$

D. $1 \oplus 0 = 1$

2. The simplified form of the Boolean expression $F = \overline{A}BC + \overline{A}B\overline{C} + \overline{A}B\overline{C} + A\overline{B}C + A\overline{B}C$ is

A. $A+B+C$

B. $A'+B'+C$

C. $A'+B'+C'$

D. ABC

3. Two 4-bit binary numbers (1011 and 1111) are applied to a 4-bit parallel adder. The carry input is 1. What are the values for the sum and carry output?

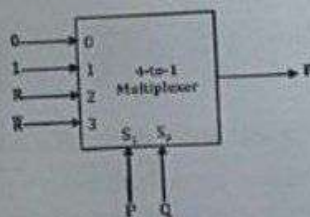
A. $\Sigma_4 \Sigma_3 \Sigma_2 \Sigma_1 = 0111, C_{out} = 0$

B. $\Sigma_4 \Sigma_3 \Sigma_2 \Sigma_1 = 1011, C_{out} = 1$

C. $\Sigma_4 \Sigma_3 \Sigma_2 \Sigma_1 = 1111, C_{out} = 1$

D. $\Sigma_4 \Sigma_3 \Sigma_2 \Sigma_1 = 1100, C_{out} = 1$

4. Consider a 4-to-1 multiplexer with two select lines S1 and S0, given below



The minimum equation of the Boolean expression for the output F of the multiplexer is

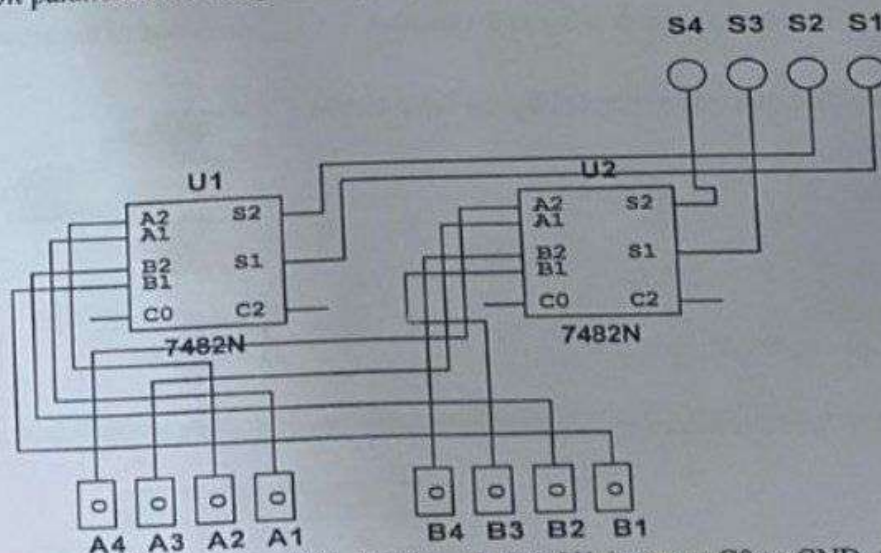
A. $P'Q + P'QR' + PQR' + PQR$

B. $P'Q + QR' + PQR$

C. $P'QR + P'QR' + QR' + PQR$

D. $P'Q' + Q'R + PQR'$

5. To implement 4 bit parallel adder using 2 ICS (7482) which is 2 bit parallel adder:



A. For U2 (connect C0 to GND, C2 to C0 of U1)

C. For U2 (connect C2 to GND, C0 to C0 of U1)

B. For U1 (connect C0 to GND, C2 to C0 of U2)

D. Answers a or b

6. The binary numbers $A = 1100$ and $B = 1001$ are applied to the inputs of a comparator. What are the output levels?

A. $A > B = 1, A < B = 0, A = B = 1$

C. $A > B = 1, A < B = 0, A = B = 0$

B. $A > B = 0, A < B = 1, A = B = 0$

D. $A > B = 0, A < B = 1, A = B = 1$

14. The following circuit compares two 2-bit binary numbers, X and Y , under what condition Z will be 1?

A. $X \neq Y$

B. $X > Y$

C. $X < Y$

D. $X = Y$



15. The following truth table represents a -----
-gate

X	Y	F
0	0	1
0	1	0
1	0	0
1	1	0

A. XNOR

B. XOR

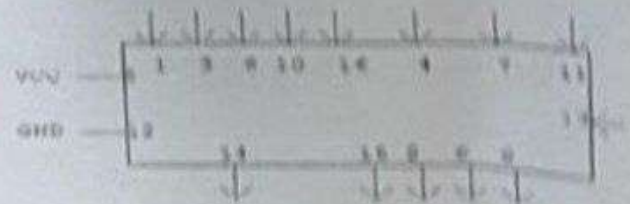
C. NAND

D. NOR

Q2: Given the following adder circuit (7 marks)

Pins 1,3,8,10 represent (input A)
Pins 4,7,11,16 represent (input B)
Pin 13 represent C_{in}
Pins 2,6,9,15 and, $S_4S_3S_2S_1$
Pin 14 represent C_{out}

Q1: the operation of this circuit is:



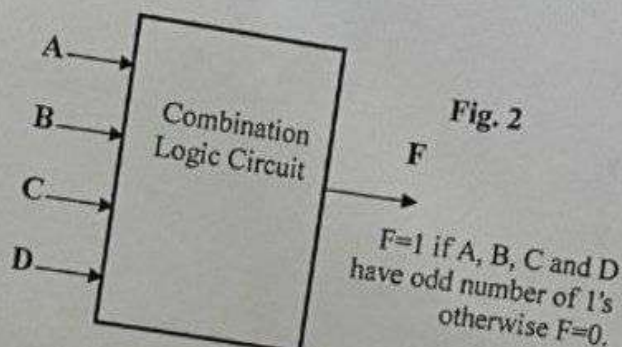
Q2: Modify the circuit above to operate as parallel adder subtractor and explain how the subtraction work?

Q3: Boolean function $F(A, B, C, D)$ is generated in a combinational circuit as shown in Fig. 2, the output is equal to "1" if the number of "1's" on input variables A, B, C, D is odd, and output F is "0" otherwise.

(6 marks)

A) Complete the following truth table?

A	B	C	D	F
0	0	0	0	
0	0	0	1	
0	0	1	0	
0	0	1	1	
0	1	0	0	
0	1	0	1	
0	1	1	0	
0	1	1	1	
1	0	0	0	
1	0	0	1	
1	0	1	0	
1	0	1	1	
1	1	0	0	
1	1	0	1	
1	1	1	0	
1	1	1	1	



B) design the circuit using basic logic gates

Q4: Build Full Adder using only 2 ICs of 4:1 MUX, using A,B as select lines. Show the truth table. (7 marks)